

Results Report

Analysis of 2012 QCEW and State Population Data

Student: Airui Meng

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This report presents cleaned and interpreted results for the 2012 QCEW and Census population data exercise. It includes data preparation notes, descriptive statistics, graphical analysis, regression results, and a discussion of data definitions and limitations.

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1 Overview

This report presents the results of a multi-part data exercise using the 2012 Quarterly Census of Employment and Wages (QCEW) county high-level files (NAICS-based) together with U.S. Census Bureau state population estimates. The objective is to transform raw administrative data into a coherent set of descriptive and regression-based findings at the state and county levels.

The report is organized to separate data preparation from substantive results. Questions 1 and 2 establish the structure of the source files and the population merge required for later analyses. Questions 3 through 7 then present the main analytical results, including summary statistics, a constructed county wage ranking, a state-level scatterplot, regression models, and a Massachusetts county comparison between annualized first-quarter wages and annual wages.

2 Data Sources and Data Preparation

2.1 QCEW 2012 High-Level Files

(The responses to the subsequent questions of Task 1)

The main annual QCEW file, `allh1cn12.xlsx`, is a 2012 employment and wage dataset organized into two sheets: one (`US_St_Cn_MSA`, 62,556 rows) covering U.S. states, counties, and metropolitan statistical areas, and the other (`US_PR_VI`, 1,585 rows) covering Puerto Rico and the U.S. Virgin Islands. Each observation represents a combination of geographic area, industry classification (NAICS code), and ownership category (e.g., private, federal, state). The file includes establishment counts, employment levels, total wages, annual average weekly wage, annual average pay, and location quotients relative to U.S..

The quarterly files, `allh1cn121.xlsx` through `allh1cn124.xlsx`, follow the same general structure but provide quarter-specific information, including quarterly wages and monthly employment measures. In this exercise, the annual file is used for Questions 3 through 6, while the first-quarter file is used for Question 7.

2.2 Census Population File

To support the state-level analyses, I used the U.S. Census Bureau file (`nst-est2018-01.xlsx`) containing annual resident population estimates for the United States, regions, states, and Puerto Rico. From this source, I extracted the 2012 state population variable and prepared a cleaned state-level population dataset.

2.3 Preparation Workflow

The data preparation process involved three main steps:

1. Cleaning the annual QCEW file and standardizing the variables needed for state-level and county-level analysis, `cleaned_data/base_qcew_annual.dta`.
2. Extracting 2012 state population values from the Census file and preparing them for a state-name merge, `cleaned_data/base_population.dta`.
3. Preparing a Massachusetts-only first-quarter QCEW subset for the annualized wage comparison in Question 7, `cleaned_data/base_qcew_q1.dta`.

3 Results

3.1 Question 3: State-Level Federal Employment Shares

To answer Question 3, I restricted the annual QCEW file to state-level observations for total employment across all industries, excluded the District of Columbia, and retained only the *Total Covered* and *Federal Government* ownership categories. I then reshaped the data so that each state had one observation and computed the federal employment share as

$$\text{Federal Share} = \frac{\text{Annual Average Employment in Federal Government}}{\text{Annual Average Employment in Total Covered}}.$$

Table 1 summarizes the main results.

Table 1. State-Level Share of Total Annual Average Employment in Federal Government, 2012

Metric	Value
Highest federal employment share	5.826% (Maryland)
Lowest federal employment share	1.076% (Wisconsin)
Mean federal employment share	2.293%
Median federal employment share	1.945%
Standard deviation	1.124 percentage points
Number of states	50

Notes. Shares are calculated as federal annual average employment divided by total covered annual average employment. The District of Columbia is excluded as required by the task.

The results show substantial variation across states in the relative importance of federal employment. Maryland has the highest federal employment share, which is plausible given its proximity to Washington, D.C. and the concentration of federal agencies and federally related work in the region. Wisconsin has the lowest share among the fifty states.

Because the mean is larger than the median, the distribution appears somewhat right-skewed, suggesting that a relatively small number of states have especially high federal employment shares.

3.2 Question 4: County-Level Mean Weekly Wage in Two Industries

For Question 4, I created a county-level dataset using the industries *Financial activities* and *Professional and business services*. Counties were retained only if both industries had non-zero and non-missing average weekly wage values. After filtering, the data were reshaped so that the two industry wage measures appeared side by side for each county, and a constructed county-level mean wage was defined as

$$\text{Mean Wage} = \frac{\text{FA} + \text{PBS}}{2}.$$

Notes. FA = Average Weekly Wage in Financial Activities; PBS = Average Weekly Wage in Professional and Business Services.

The final analytic dataset contained 2,878 counties. Table 2 reports the five counties with the highest mean weekly wage under this construction.

Table 2. Top 5 Counties by Mean Weekly Wage Across Financial Activities and Professional and Business Services, 2012

Rank	County	State	Mean Weekly Wage (\$)
1	San Mateo County	California	3,599.00
2	New York County	New York	3,486.50
3	Fairfield County	Connecticut	3,297.00
4	San Francisco County	California	2,747.00
5	Suffolk County	Massachusetts	2,574.00

Notes. Only counties with non-zero wage values in both industries are included. The reported figure is the simple arithmetic mean of the two industry-level average weekly wages.

This ranking is dominated by counties associated with major financial and professional-service centers. These are major financial centers where high-skill service employment is concentrated.

3.3 Question 5: State-Level Relationship Between Employment Share and Annual Pay in Education and Health Services

For Question 5, I restricted the annual QCEW file, to state-level observations in the *Education and health services* sector and merged those observations with the 2012 state population data. I then defined the employment share as

$$\text{Employment Share} = \frac{\text{Average Employment in Education and Health Services}}{\text{State Population}},$$

and converted the measure to percentage form for presentation. The merged dataset contains 51 observations, corresponding to the 50 states plus the District of Columbia.

Figure 1 displays the state-level relationship between the share of the population employed in Education and health services and annual average pay in that sector.

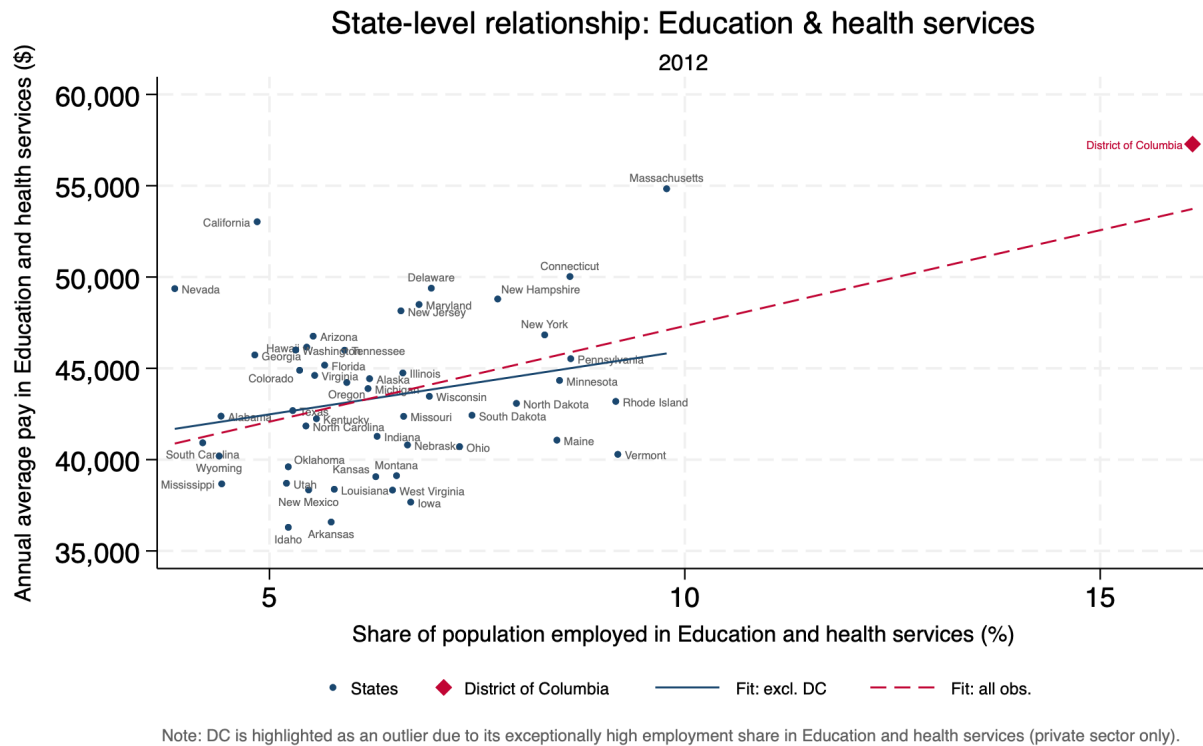


Figure 1. State-Level Relationship Between Employment Share and Annual Average Pay in Education and Health Services, 2012

The figure shows a clear positive association: states with a larger share of their population employed in Education and health services tend to exhibit higher annual average pay in that sector. At the same time, the scatters are not perfectly tight, and there is visible dispersion around the fitted line. This indicates that employment share alone does not fully explain cross-state pay differences.

The District of Columbia is highlighted separately (red diamond) because it is a clear extreme outlier, with an employment share of approximately 16% and a salary of \$57,280—far from the range of any state. To illustrate DC’s influence, Figure 1 includes two fitted lines: the solid line is estimated on the 50 states only, while the dashed line includes DC. The visible gap between the two lines confirms that DC exerts substantial leverage on the estimated slope.

An important data-definition note applies here. In the QCEW high-level county files, supersectors such as *Education and health services* are recorded only under private ownership rather than total covered ownership. As a result, the employment share used in this figure reflects private-sector employment in Education and health services relative to total state population, not total sector

employment across all ownership categories. This does not invalidate the analysis, but it affects how the relationship should be interpreted.

3.4 Question 6: Regression-Based Analysis of Annual Pay, Employment Share, and Population Size

Using the state-level dataset constructed in Question 5, I estimated a sequence of linear regression models to describe the relationship between annual average pay in Education and health services, the sector's employment share, and state population size.

The dependent variable is annual average pay in Education and health services. The main explanatory variable is the percentage of the population employed in that sector, and state population size is represented by the natural logarithm of 2012 population.

I estimated four specifications:

1. a simple bivariate model;
2. a multiple regression that adds log population;
3. an interaction model including employment share $\times$ $\ln(\text{population})$
4. a centered interaction model to make the main effects easier to interpret.

Table 3 presents the regression results.

The simple model in column (1) indicates a positive and statistically significant association between employment share and annual average pay. Interpreted directly, a one-percentage-point increase in the share of the population employed in Education and health services is associated with approximately \$1,049 higher annual average pay in that sector.

Once state population size is added in column (2), the estimated association becomes even larger: a one-percentage-point increase in employment share is associated with roughly \$1,283 higher annual pay, holding log population constant. Log population is itself positive and statistically significant, suggesting that more populous states also tend to have higher annual pay in Education and health services. The R^2 rises from 0.205 to 0.318, indicating improved explanatory power after accounting for population size.

Column (3) introduces an interaction between employment share and log population to examine whether the relationship between employment share in education and health services and annual average pay depends on state population size. Although the interaction coefficient is negative, it is not statistically significant. Therefore, the relationship between employment share and pay does not appear to depend on state size. Because interaction models are often easier to interpret after centering, column (4) re-estimates the model using centered variables. The centered specification leads to the same substantive conclusion: both centered employment share and centered log

Table 3. Regression Results: Determinants of Education and Health Services Annual Average Pay

Variable	(1) Simple	(2) Multiple	(3) Interaction	(4) Centered Interaction
Employment share (%)	1048.66*** (315.74)	1282.83*** (326.12)	5674.33 (3644.56)	
ln(Population)		1531.10** (587.15)	3575.02* (1893.08)	
Employment share × ln(Population)			-307.93 (263.80)	
Centered employment share (%)				1014.82** (438.20)
Centered ln(Population)				1560.68*** (576.83)
Centered interaction				-307.93 (263.80)
Constant	36834.08*** (2167.89)	12134.11 (9814.12)	-17217.93 (27021.08)	43516.53*** (570.69)
Observations	51	51	51	51
R^2	0.205	0.318	0.340	0.340

Notes. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Model (4) centers both employment share and ln(Population) before constructing the interaction term.

population are positively associated with annual pay, while the interaction remains statistically insignificant.

As noted in Question 5, DC is an extreme outlier with an employment share roughly 6 percentage points above the next highest state. With only 51 observations, a single high-leverage point can materially shift the estimated coefficients. Table 4 presents the results of re-estimating the simple and multiple regression models after excluding DC.

Table 4. Robustness Check: Comparing Results With and Without DC

Variable	(1) Simple All obs.	(5) Simple Excl. DC	(6) Multiple Excl. DC
Employment share (%)	1048.66*** (315.74)	697.38 (503.90)	891.11* (455.82)
ln(Population)			1580.65*** (563.93)
Constant	36834.08*** (2167.89)	38993.73*** (3242.79)	13789.73 (9749.95)
Observations	51	50	50
R^2	0.205	0.057	0.204

Notes. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$. Column (1) reproduces the simple model from Table 3 for comparison. Columns (5) and (6) exclude the District of Columbia.

The comparison is instructive. Removing DC causes the employment-share coefficient in the simple model to fall from 1,049 to 697 and to lose statistical significance entirely, while R^2 drops from 0.205 to 0.057. This confirms that DC—as an extreme outlier in both employment share and pay—exerts substantial leverage on the bivariate relationship.

Once log population is added (column 6), the employment-share coefficient partially recovers to 891 and becomes marginally significant ($p \approx 0.057$), and log population itself remains strongly significant. The R^2 of the multiple model excluding DC (0.204) is comparable to that of the simple model with DC (0.205), suggesting that much of the apparent explanatory power in the bivariate specification was driven by a single observation rather than a genuine cross-state pattern.

Overall, the regression evidence supports a positive descriptive relationship between sector employment share and annual pay, along with a positive association between population size and annual pay. However, the robustness check demonstrates that these associations are sensitive to the inclusion of DC, and the results should be interpreted as cross-state associations rather than causal effects.

3.5 Question 7: Massachusetts Counties with Annualized Q1 Wages Greater Than Annual Wages

For Question 7, I combined the first-quarter QCEW file, with the annual QCEW file. I restricted both datasets to Massachusetts counties, retained only *Total, all industries* and *Total Covered* ownership, excluded *Unknown Or Undefined, Massachusetts*, annualized first-quarter wages by multiplying first-quarter total wages by four, and compared that value with annual total wages.

After merging the annual and first-quarter county datasets, all 14 Massachusetts counties matched successfully. Of those 14 counties, 2 had annualized first-quarter wages greater than full-year annual wages. The counties are shown in Table 5.

Table 5. Massachusetts Counties with Annualized Q1 Wages Greater Than Annual Wages, 2012

County	Annualized Q1 Wages (\$)	Annual Wages (\$)
Middlesex County, Massachusetts	61,559,496,704	59,617,851,335
Suffolk County, Massachusetts	52,076,920,832	48,081,832,775

Notes. The comparison uses *Total Covered* ownership and *Total, all industries*. *Unknown Or Undefined, Massachusetts* is excluded. Annualized Q1 wages are defined as four times first-quarter total wages.

These results indicate that Middlesex and Suffolk Counties had first-quarter wage totals so high that multiplying those values by four produced figures larger than their annual total wages.

4 Overall Conclusion

This exercise uses publicly available QCEW and Census data to produce state- and county-level findings. The QCEW files support several kinds of analysis, ranging from descriptive summaries to more structured regression-based approaches.

Three findings are worth noting. First, federal employment shares vary substantially across states, with Maryland at the top and Wisconsin at the bottom among the fifty states. Second, counties associated with major metropolitan service economies dominate the wage ranking constructed from Financial activities and Professional and business services. Third, in Education and health services, both the graphical evidence and the regression results indicate a positive association between employment share and annual average pay, while larger state population is also associated with higher pay.

The Massachusetts county comparison in Question 7 identifies two counties whose annualized first-quarter wages exceed annual wages, showing that quarterly records sometimes diverge from annual totals in ways worth investigating.

5 Limitations and Data Notes

Several data considerations are important for interpreting these results correctly.

First, in the QCEW high-level files, supersectors such as *Education and health services*, *Financial activities*, and *Professional and business services* are reported only under private ownership rather than total covered ownership. Consequently, the employment share used in Questions 5 and 6 reflects private employment in the sector relative to total state population rather than total sector employment across all ownership categories.

Second, the wage ranking in Question 4 is based on a constructed county-level mean of two industries. It is useful for comparison, but it is not itself an official published wage measure.

Third, the regression analysis in Question 6 is descriptive. The coefficients summarize cross-state associations and should not be interpreted as causal estimates.

Fourth, in Questions 5 and 6, the sample size is only 51 (or 50 excluding DC). As the robustness check in Table 4 demonstrates, a single extreme outlier such as DC can materially shift coefficients and significance levels. Results should be read with this sensitivity in mind.

Finally, the Question 7 comparison identifies counties with unusual wage patterns, but it does not by itself establish why those patterns occur.